

Cruise Outbreak Computation and Publication Program

AWS-resident evidence, unattended control,
and a reusable eight-paper research program

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Confidential

Executive Summary

Answer 1: connect the control path, not the raw-data path. Protected short GitHub dispatch workflows exchange an OpenID Connect (OIDC) token for a scoped Amazon Web Services (AWS) role and wrap existing Batch submission scripts. Simulation, reduction and Stan inference stay beside Amazon Simple Storage Service (S3) data. Separate short GitHub COLLECTOR workflows read durable S3 receipts/results and publish a bounded index through GitHub's application programming interface (API), using scoped `GITHUB_TOKEN`. OIDC grants GitHub an AWS role, not AWS a GitHub credential. This is proposed, **not deployed**; Section 2 defines the authorization. Edison: `76fa2216`

Build on existing code. Selected norovirus scanners already read ZIP members by S3 byte range. Boundary and Sentinel analysis entrypoints already support AWS-side analysis. The missing work is durable orchestration, validated bounded reducers, runtime provenance, secure admission and reliable result consumption—not a replacement simulator or a new bulk-transfer service. [1] Edison: `5d4be453`

Answer 2: organize eight claim-driven paper programs and a reusable software companion. Start with a coherent, bounded methods/validation narrative, then diagnostics, wearables, wastewater, response, heating, ventilation and air conditioning (HVAC), fleet–port inference and variant detection. Retain vessel-specific outcomes and add explicitly targeted fleet standardization and heterogeneity. Neither layer replaces the other. Reuse qualified runs; request additional computation only when it resolves an unanswered decision or precision requirement. Edison: `8171e84c`

Current evidence, not September readiness. This report uses the supplied canonical source at `ca721b871dd19ed903165bc1a4b6d06cff1f9def`, dated 18 October 2026. Current ledgers contain both measurements and changing status. In particular, norovirus mega wave 3 is documented as submitted, awaiting cells—not completed. A failed security quality gate remains failed pending adjudication. The review is bounded source synthesis, not independent model revalidation. Edison: `f85969ba`

Immediate commitment: qualify the smallest useful dispatch/collection path while authors consolidate qualified evidence and draft the first manuscript. Do not start bulk campaigns, deploy infrastructure or spend from this report. One-time authorization covers GitHub permissions, AWS roles and budget policy. Unattended Edison re-entry is a separate unverified gate, not a prerequisite to evidence consolidation or writing. Humans own scientific scope, budget exceptions and release; they do not shuttle routine commands or files.

Decision boundary: AWS job completion alone does not establish successful reduction, scientific adequacy, publication readiness or a guaranteed Edison callback. The implementation and paper gates below keep those states separate.

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1 Current evidence and material reconciliation

The evidence envelope is the canonical `main` snapshot of 18 October 2026, not an old local checkout. Source inspection, repository-recorded measurements, API documentation and recommendations are distinguished throughout. Earlier engine SHAs remain attached to their own results; reading a ledger at the current SHA does not rerun those experiments. Historical reports are preserved. Only the statements in Table 1 are superseded or qualified here. [1] Edison: `5d4be453` Edison: `f85969ba`

Table 1: Material corrections and their downstream consequences.

Earlier implication	Controlling current reading	Consequence
Mega wave 3 not submitted	Later dated parent ledger: submitted 18 October, awaiting cells; older headers conflict. [2]	Do not relaunch it or claim a completed result.
Wave 2 still in flight	Parent ledger records a 17 October readout. [2]	Reuse recorded evidence after qualification; not a fresh rerun.
Security failure is external drift	27 checks: 26 success, one failed SonarCloud quality gate; cause unadjudicated. [3]	Failed remains failed; investigate before release.
Three range requests / measured savings	One HEAD + four ranged GETs on the successful path; sizes in source are illustrative.	No throughput or 260 GB decomposition inferred.
All readouts need bulk transfer	Some norovirus scans are S3-native; boundary/Sentinel analysis already runs in AWS.	Extend existing paths; qualify remaining local readers.
260 GB derived from seed counts	Volume is user-reported; no S3 inventory or transfer history verified.	No campaign-size decomposition or measured savings claim.
Small wrapper completes automation	Disk bounds, distributed admission, provenance and callback still need qualification.	No line-count delivery promises or turnkey claim.
Current SHA supersedes every old audit	Only identified statements are corrected; historical engine evidence retains its own SHA.	Target-specific reuse/bridging, not blanket replacement.

Scientific interpretation. Norovirus provisioned-source evidence provides a candidate three-hull mechanism narrative, not independently measured contamination prevalence. The mega extension needs its actual completed readout. COVID-19 observation/crew/timing discrepancies and influenza checks must be assessed by outcome and measurement version; neither a September arm-status page nor successful execution supplies a universal present-day diagnosis. Influenza's README explicitly describes internal checks without a train/test split. [2, 4–6]

The commissioned-flow ledger supports a narrow engineering verification result for two hulls. Agreement with declared flow fields is not independent field validation of all vessels, nor proof of pathogen-specific infection effects. Preserve that distinction for P6. [7]

The full publication provenance envelope is still documented as proposed and not wired into execution. Existing hashes, manifests and ledgers are useful evidence, but do not imply complete runtime run-to-claim linkage. [8]

Practical consequence: qualify the specific pathway supporting each claim. Do not restart measured campaigns because a blanket readiness label is stale; do not promote a submitted campaign to evidence because its manifest exists. Archived source evidence and the reconciled findings are linked in Data sources. Pre-correction executive summaries, line-numbered reports and hash lists remain historical derivatives, not controlling current conclusions.

2 Recommended asynchronous architecture

The text diagram below is the proposed control/data separation. Its end-to-end operation remains unqualified. Official dispatch and federation interfaces support the design; the account configuration and Edison handoff still require implementation and testing. [9, 10] Edison: `76fa2216`

```
Edison: frozen manifest + allowlisted scientific request
  -> protected short GitHub DISPATCH -> OIDC AWS submit role
  -> admission + S3 attempt records -> existing Batch scripts
  -> S3 raw/checkpoints -> bounded reducers / Stan
  -> S3 durable receipts + compact validated results/index
GitHub COLLECTOR: scheduled/on-demand short workflow
  -> OIDC AWS read-only role -> inspect S3 receipts/results
  -> scoped GITHUB_TOKEN -> bounded GitHub API summary
Edison callback/re-entry: separate, unverified gate
No runner stays alive for the whole campaign.
```

2.1 A request is a scientific contract

Accept a reviewed manifest reference and hash, not arbitrary shell commands. Identify project, campaign, attempt and parent; exploratory/confirmatory/revision purpose; engine SHA, controller SHA and image digest separately; resolved configuration, inputs and versions; effective seeds; vessel/role populations; fleet target and weights where relevant; estimands, plan and stop rule; analysis version; output/telemetry contract; approvals and resource ceilings. Freeze the attempt before admission. New analysis can reuse old raw data under a new analysis ID without pretending the engine was rerun.

2.2 Durable execution and return

A short protected Actions job obtains temporary AWS credentials and invokes existing submission scripts after admission. Durable S3 attempt records retain the decision, fingerprint, job IDs and lifecycle state; existing workers/reducers write receipts and results. Use Batch dependencies where appropriate, but verify the complete expected shard set before inference: a dependency is not scientific validation. Reconciliation can run in bounded scheduled jobs rather than a custom always-on service. The read-only collector does not admit work, schedule reducers or mutate AWS state; any such action uses a separately authorized dispatch/reconciliation role. Simulation, rescoring, posterior predictive checks and deep diagnostics remain in AWS. [11–13]

Authorized reverse publisher: the GitHub COLLECTOR workflow. In lean v1, short scheduled or on-demand protected collection workflows assume an AWS read-only role through OIDC, inspect durable S3 receipts and compact results, then publish an API-readable bounded index, check or issue summary using their own scoped `GITHUB_TOKEN`. OIDC authorizes GitHub to assume an AWS role; it does **not** give AWS a GitHub credential. Grant only the selected publishing permission, for example `issues: write` for an issue summary or `checks: write` for a check, plus `id-token: write` for OIDC and any necessary repository read. Test the actual API representation and repository policy. Never keep a runner alive for the campaign or put credentials in public logs. [10, 14, 15]

Publish a deliberately compact index: attempt/fingerprint, lifecycle state, expected/completed/failed/duplicate counts, source and reducer digests, target/denominators, adequacy flags, artifact hashes/locations, scan/export bytes and accounting. A successful submission workflow means **submitted**, not scientifically complete. Suggested states are proposed, approved, admitted, running, reducing, validating and published, with blocked, failed, cancelled and quarantined branches. Bound each collection pass by entries, bytes and duration; resume from durable attempt IDs rather than polling inside one long-lived runner. Scheduled workflows can be delayed or disabled; retain results in S3 and make collection retryable, with on-demand recovery and visible staleness. [15]

An optional later completion bridge may use a GitHub App with an installation-scoped, short-lived token and the minimum repository permissions. AWS-side code can publish or request collection only through that separately authorized App identity, not through OIDC alone. Keep App secrets and tokens in approved secret storage, never in public logs. This bridge is not required for v1 and does not establish Edison callback/re-entry. [16]

GitHub artifact downloads can redirect to another host. Provide compact status and paginated result rows through a tested GitHub API-readable representation, such as a dedicated result-index location accessible through the selected integration. Do not require artifact ZIP download to recover the receipt. Choose access controls, size limits and retention before implementation; never place raw campaigns or secret-bearing links in public logs. [17]

2.3 Closing the Edison loop is a separate gate

Durable S3 receipts survive the initiating chat and runner; short GitHub collectors can publish their results. Neither this collector nor an optional GitHub App completion bridge guarantees Edison re-entry. A separately authorized callback identity or persistent polling service must enter the intended Edison project, retrieve the compact index and acknowledge the attempt. Existing chat-message API surfaces are not a proven external service. Best-effort wake-ups or polling only while a turn is alive do not satisfy unattended operation. If this handoff fails, keep results durably available and mark autonomous consumption **blocked**, not "done"; qualified evidence consolidation and manuscript drafting may still proceed. Edison: `76fa2216`

3 Keep raw evidence in AWS without unbounded staging

There is already useful implementation to extend. The campaign worker uploads each child artifact to S3. The boundary entrypoint provides bundle, surface, Stan, Monte Carlo and report phases; Sentinel recovery also has AWS-side fit/score machinery. These are source-supported capabilities, not evidence that this report deployed or executed them. [18–20] Edison: `5d4be453`

3.1 Selective reads are real, but format-specific

For a successful stored/deflated member, the norovirus helper performs one HEAD and **four** ranged GETs: archive tail, central directory, local-header probe and compressed member. Its body-byte accounting is

$$B_{read} = \min(65{,}536, B_{object}) + B_{directory} + 31 + B_{member}.$$

This includes overlapping bytes; request overhead, listing and retries are additional. A small object can be read in full by the tail request. The code is not a general ZIP64 reader. Before reuse across tool families, qualify ordinary and ZIP64 archives, fused versus per-run layouts, missing/corrupt members, compression methods, checksums, range failures, resource limits and restart behavior. Unsupported formats must fail clearly or enter a bounded alternative path. Some scanner modes also fetch census members; the helper does not share directory reads across invocations. The corrected offline probes establish request semantics, not live S3 throughput. [21] Edison: `f85969ba`

3.2 Reduction contract

Moving `s3 sync` into an undersized Batch job only moves the disk failure. Inventory object count, compressed/expanded sizes and required members first. Process bounded shards; stream or selectively fetch members, update incremental aggregates, checkpoint progress and release temporary storage. Stage complete archives only when a necessary computation requires them and the measured disk/memory envelope permits it. Some existing entrypoints download prefixes; audit each phase rather than labeling all AWS-resident code "streaming." [19, 20]

Retain raw data with an explicit access/retention policy. Materialize typed, versioned derived tables for common analyses; partition and columnar formats can make selective queries cheaper, but actual scan savings require measurement. A reducer must identify its estimands, units, denominator rules, selection, uncertainty method and telemetry sufficiency. If a changed likelihood needs individual timing or trajectories, rerun analysis against raw data in AWS. A compact table is sufficient only for the questions it preserves. [22, 23] Edison: `8171e84c`

Record **four separate quantities**: raw bytes stored; S3 bytes read/scanned and requests; peak AWS disk/memory plus compute time; and bytes exported outside AWS. A full AWS scan followed by a small export can solve human transfer while still incurring substantial AWS cost.

Illustration, not benchmark. With the user's decimal 260 GB planning assumption, a 0.5 GB one-way compact return reduces exported bytes by $100(1 - 0.5/260) = 99.81\%$ versus one raw download. A raw download plus reupload would be 520 GB. Neither arithmetic result estimates scan cost, runtime, scientific sufficiency or achieved production performance. Edison: `76fa2216`

Core exports should preserve all-voyage outcomes, extinction/no-detection, role denominators, seed/block completeness and diagnostics. Add a justified full-telemetry validation panel and targeted tail cases; do not delete inconvenient failures or replace the archive with a summary percentage.

4 Authorization, admission and integration status

Use the existing GitHub connection as the first control path, not as evidence of AWS access. Current supplied checks show `/user` and workflow-list reads working; `actions/permissions` GET returned 403. These observations do not prove or disprove dispatch authorization. **No live dispatch was performed.** Fine-grained personal access token (PAT) Actions write, repository selection and the chosen API response contract must be tested once in a non-spending handshake. Configure OIDC trust to the reviewed repository/environment and audience; separate build, admission and job roles. [9, 10]

A later narrow AWS API is optional, not a prerequisite. AWS Identity and Access Management (IAM) authorization requires Signature Version 4 (SigV4); a configured JSON Web Token authorizer is a different authentication scheme. A static bearer token does not become SigV4. Never paste credentials into chat or export runtime credentials to a callback worker. Use approved secure integration and secret-storage channels. [24, 25]

4.1 Admission and failure accounting

Reserve capacity/budget transactionally before dispatch; enforce allowed image/config/analysis, maximum concurrent work, attempts, runtime, output and query scope. Replaying an identical attempt returns its receipt; changed content under the same ID is rejected. Ordinary Batch `SubmitJob` has no documented `clientToken` in the reviewed interface. Cover submission/ledger crash windows through a durable outbox/reconciler, correlated job tags and explicit duplicate/ambiguous-attempt handling; do not claim exact-once submission from a local dictionary test. [11] These are logical admission, cost and duplicate-control requirements, not a mandate for a new orchestration stack. Select the smallest existing implementation that satisfies them; durable S3 records alone are not proof of atomic reservation or safe submission. A serialized protected dispatcher with durable state may suffice at limited scale only after its concurrency and crash tests pass; add a managed transactional component if required.

Budget alerts are **not a hard cap**. AWS billing updates lag; some Batch allocation strategies can exceed `maxvCpus` by one instance. Use conservative reservations informed by measured pilots, bound retries and queued work, stop admission on thresholds, cancel/terminate according to policy and reconcile actual usage. Include storage, scans/requests, inference, logs and export—not only simulation CPU. No dollar budget or campaign size is inferred from 260 GB. [26, 27]

Table 2: Integration evidence and the remaining gate. MCP: Model Context Protocol.

Service	Observed status	Needed before unattended use
GitHub / Drive	GitHub read verified as described; Drive reported connected, untested here.	GitHub dispatch/readback test; protected workflow + OIDC once. Drive optional, never raw-data relay.
AWS / Edison callback	AWS topology/identity and external callback unverified; operational loop blocked.	Account owner authorizes roles/budget; integration owner proves durable re-entry.
Consensus	Public API/MCP documented; no attached connection. [28]	Secure access, terms and output provenance; otherwise use existing cited sources, without claiming automated search.
Devin / Cursor	APIs documented; session authorization unknown. [29, 30]	Approved implementation workflow and reviewed pull requests; no autonomous launch assumed.
SonarQube / CodeQL	Check evidence exists; one security gate failed. Extra API/entitlement status unknown.	CI owner investigates finding and verifies actual scan outputs; missing checks are not passes.
Refine / Manusights	Interfaces documented; credentials, confidentiality and end-to-end use unverified. [31, 32]	Author approves manuscript release and integration; if unavailable, record blocked service and use independent review fallback.

Table 2 separates a documented interface from a connected, authorized and tested integration. Optional vendor reviews cannot silently become mandatory blockers for every scientific result. If selected, the owner records trigger, permitted manuscript/code input, confidentiality, expected output, cost/retries and fallback. Humans approve science, spend and release—not routine tool relay. Edison: `76fa2216`

5 Implementation stages and the first qualification

Minimum useful deployment: wrap the current Batch/S3 submission and reduction scripts with protected dispatch and short collection workflows, plus durable S3 attempt/index records and qualified admission, cost and duplicate controls. No new API Gateway, full Step Functions deployment, Athena migration or custom always-on control service is a prerequisite for v1. Reuse the smallest existing implementation that passes the controls; a managed controller is optional later for scale. This is a proposed integration of inspected code, not a claim that the wrappers or controls already pass. [1]

Table 3 is a proposed work sequence, not authorized deployment. Effort bands are engineering planning judgments: **small** means a few focused working days for one experienced owner once access is ready; **medium** means roughly one to several engineer-weeks; **large/uncertain** spans multiple weeks or an unresolved service dependency. They are not quotations, measured delivery times or a summed schedule. Identity approvals, existing AWS topology and reducer complexity can dominate elapsed time. Stages are workstreams, not a freeze on writing until automation is fully unattended.

Table 3: Proposed implementation stages, deliverables and acceptance.

Stage / owner / effort	Deliverable	Acceptance and stop
S0 Scope / PI + AWS owner / small	Topology map, claim card, existing-run inventory, rights and budget policy.	Target, data sufficiency and owner approved; missing account/science facts block admission.
S1 Identity / integration owner / medium	Protected dispatch and short collector; separate AWS submit/read roles and scoped GitHub publishing token.	Non-spending dispatch/collection readback and denial tests; bounded API output, no credential leakage.
S2 Control / cloud engineer / medium–large	Existing Batch/S3 scripts plus durable attempt/index records; minimal reservation, duplicate and cancellation controls.	Replay/crash/duplicate/timeout tests; ambiguous job state visible, not silently accepted.
S3 Data / analysis engineer / medium–large	Reuse range reader; bounded reducers; provenance and summary adequacy validation.	Known small data reproduce expected results within resource bounds; unsupported formats fail clearly.
S4 Loop / integration + science owners / large uncertainty	Qualify dispatch/collection first; separately qualify durable Edison callback/polling acknowledgement.	Collection survives runner exit; resources/results reconcile. Unattended Edison claim blocked until re-entry passes.
S5 Release / model lead + author / medium, then iterative	P1 consolidation/drafting in parallel; evidence capsule, install test, claim graph and pilot-informed next plan.	Claim-specific evidence and author release approval; no bulk sizing before bounded pilot/control evidence.

5.1 Minimal first vertical slice

Select **one already qualified, small scientific input/fixture** with a known expected result—not a new fleet-sized campaign. Freeze its source, container, manifest and expected denominators. Start with a no-spend control handshake, then a separately approved bounded AWS pilot. Reuse a small existing raw object set if that is sufficient to qualify the reducer; do not relaunch an ongoing scientific wave. Edison: `76fa2216` Edison: `8171e84c`

The minimum useful slice must demonstrate request receipt; admission or explicit denial; bounded job and output; verified S3 inventory; reducer equivalence against known expected outputs; authorized collector publication with compact API-readable status/results; and scientific adequacy flags. Measure queue/runtime, scan/read/export bytes, peak disk/memory and actual accounting before sizing anything larger. Durable acknowledgement in Edison is an additional acceptance gate for the fully unattended loop, not for starting qualified evidence consolidation or drafting.

Required fault tests: replay the same request; alter its fingerprint; exhaust reservation; interrupt the worker; expire credentials; lose/duplicate completion events; remove or corrupt a shard; reject an unsupported ZIP format; mismatch the reducer schema; cancel; and make artifact redirects inaccessible. Confirm failures are visible, retries are not counted as independent replicates, incomplete results are not published and ambiguous submissions do not silently duplicate accepted work.

The decisive unattended test: end the initiating chat/runner, publish a synthetic terminal receipt, trigger collection, and prove an authorized callback or durable poller re-enters the correct project, consumes the compact result once and stores an acknowledgement. Retry a lost acknowledgement and recover by attempt ID. If only a human can wake each campaign or relay its output, the requested integration has not passed.

Advance new computation only when its control path, bounded data path, scientific output contract and human scope/budget approval pass. Do not freeze all work pending unattended Edison re-entry: authors can consolidate existing qualified evidence, prepare admissibility cards and draft manuscripts while automation setup proceeds. Scientific release remains claim-specific. Repeat the relevant slice when a load-bearing interface changes, not mechanically for unchanged qualified analyses. A later performance pilot should resolve capacity uncertainty, not justify bulk exploration by default.

6 Scientific scope: vessel and fleet evidence together

P1 should begin with the strongest coherent methods/validation narrative, not universal validation or a monograph. A bounded mechanism or documented failure can be scientifically useful if its claim, observation process and validity domain are explicit. Current ledgers support evidence consolidation; they do not independently certify every pathogen, hull or outcome. Edison: `8171e84c`

6.1 Disease-specific admissibility cards

For every primary claim, record pathogen/strain and mechanism version; geometry and realized complement; passenger/crew split; itinerary/era and operations; estimand and observation channel; calibration sources and historical tuning; independent validation; discrepancy; supported range and excluded extrapolations; evidence owner/date and release status. Keep **hull, fleet frame, total population and crew/passenger composition** as four separate axes. An axis can be not applicable with a claim-specific reason; missing evidence for a load-bearing axis blocks that claim, not automatically every module.

6.2 Retain vessel outcomes; standardize deliberately

For each vessel/voyage retain infection, illness, care-seeking, specimen, confirmation and posting counts separately, with role-specific denominators and uncertainty. Report all-voyage results as well as outbreak-conditioned results. Extinction is a valid outcome. Match exogenous introductions across policy comparisons where possible; the same numeric seed alone does not prove the same causal exposure history after an engine change.

For a declared target with effects Δ_h and weights w_h, a proposed standardized contrast is $\Delta_F = \sum_h w_h \Delta_h$, with nonnegative weights summing to one. State whether weights represent vessels, voyages, passenger-voyages or person-time. For voyage-frequency weights, a population attack rate can be $\sum_h w_h E[C_h] / \sum_h w_h N_h$; it is not generally the mean vessel attack rate. Define ratio-of-expectations versus expected-ratio when voyage denominators vary. Preserve passenger and crew results alongside the total. Edison: `8171e84c`

A named, dated fleet estimate requires a defensible traffic frame. Equal-weighted hull templates yield a **design-set summary**, not all-cruise incidence. When fleet weights are unknown, retain vessel effects and scenario-weighted envelopes. Propagate shared parameter draws jointly, including their cross-vessel covariance; add weight, measurement and structural uncertainty. Partial pooling requires defensible exchangeability. More seeds on four archetypes do not create more independent real vessels.

6.3 Selection and response are part of the model

The Centers for Disease Control and Prevention Vessel Sanitation Program (VSP) concerns acute gastroenteritis (AGE), not generic respiratory surveillance. Its monitoring guidance distinguishes reporting at 2% from an outbreak threshold of 3% of passengers **or** crew. Posting-conditioned data need a selection model; posting frequency needs an eligible voyage denominator including non-posted voyages. Do not infer unconditional infection incidence from posted outbreaks alone. [33]

Response changes both infection and ascertainment: more confirmed cases can accompany fewer infections. Freeze the observed-information rule and capacity in causal comparisons. Keep calibration, internal consistency, synthetic recovery and independent validation distinct; a newly assigned holdout cannot erase past tuning. Hold out independent events/vessels/periods when available, report source overlap, and compare predictive skill with simpler baselines. Diagnose parameter confounding before interpreting fitted constants.

7 Prioritized publication matrix: P1–P4

Tables 4 and 5 define **eight proposed paper programs**, not eight required campaigns or promises of novelty/acceptance. Priority reflects dependencies; work can proceed in parallel. Every program preserves a vessel result and a fleet layer. The latter can initially be a clearly labeled design-set standardization while a named-fleet frame is developed. The shared software companion is part of P1's release strategy, not a ninth substantive study. Edison: `8171e84c`

Table 4: Publication matrix, priorities P1–P4. All outcomes and gates are proposed.

Program / question	Estimand and contrast	Vessel + fleet outputs	Minimum evidence / next step
P1 Model validation + software	Held-out predictive skill across selected pathogen/hull/era targets versus simpler baselines.	Vessel predictive distributions and discrepancies; declared weighted mixture and heterogeneity.	Freeze coherent core; distinguish calibration/internal checks from independent validation; package software.
P2 Diagnostics	Detection delay, false isolation and net health benefit of test/retest rules at equal capacity.	Role-specific timeliness/burden; standardized benefit with access/capacity modifiers.	Assay and specimen/presentation timing, queues and response verified; freeze one informative contrast.
P3 Wearables	Intention-to-offer benefit, alert burden and interception: preboarding versus onboard.	Vessel uptake/role decision curves; uptake–risk weighted fleet effects.	Confounder-aware signal performance, voluntary selection and downstream action; bound missing real-world inputs.
P4 Wastewater	Detection probability/lead time versus sampling, mixing and assay loss at fixed response.	Tank/zone signal and false negatives; comparable-plumbing fleet alert workload.	Mass/units and assay calibration, shedding/decay uncertainty; first identify limiting mechanism.

7.1 P1: make the first paper coherent and reusable

Select one connected set of qualified mechanisms and outcomes, explain what the model can and cannot predict, and compare its performance against independent targets and simpler alternatives. Current norovirus cross-hull development, observation/response tensions and narrow airflow verification are candidate evidence—not a mandate to combine every subsystem. Preserve failed mechanisms and negative results where they explain model choice. If independent validation is unavailable, present a verification/structural-diagnosis paper with a narrower title and claim, rather than relabel calibration as validation.

Use Overview, Design concepts and Details (ODD) for a reproducible model description; TRACE for rationale, development, testing and intended use. These reporting frameworks help a reader assess the evidence but do not validate the simulator themselves. [34, 35]

7.2 P2–P4: isolate measurement from intervention

For diagnostics, compare test/retest policies at equal capacity and fixed response rules, separating ascertainment gain from infections averted. For wearables, distinguish voluntary preboarding selection from onboard monitoring; vary uptake–risk association and confounder correction. An anomalous signal is not a pathogen diagnosis. For wastewater, distinguish shedding, plumbing, dilution, extraction and assay detection; gene signal is not an infected-person count.

Each program first needs independent evidence for its load-bearing measurement layer, followed by a frozen policy contrast. Where such evidence is missing, bounded synthetic trade-offs can be informative, but absolute operational lead times or clinical effectiveness remain gated. The existing source includes wearable and Sentinel design material; specifications are not completed validation. [36–38]

8 Prioritized publication matrix: P5–P8

Table 5: Publication matrix, priorities P5–P8. Local gates do not require waiting for P1 publication.

Program / question	Estimand and contrast	Vessel + fleet outputs	Minimum evidence / next step
P5 Clinical/public-health response	True infections/illness and burden under paired policies at matched observed information.	Time-resolved effects and role harms; population-weighted benefit and policy heterogeneity.	Response-state/event-order integrity, capacity and service feedback; test ranking-changing regimes.
P6 HVAC	Exposure/infection changes under flow/filtration/occupancy contrasts with equivalent resources.	Commissioned transport/exposure by zone; conditional cross-hull standardization.	Commission each claimed hull; distinguish design-flow verification from field validation and dose-response.
P7 Fleet–port inference	Identifiable port/week hazard contrasts and warning value against surveillance baselines.	Voyage likelihood/source checks; identified fleet contrasts, coverage and vessel residuals.	Rank/connectivity, source-market separation and synthetic recovery before large inference.
P8 Variant detection	Correct detection latency, false novelty and source ambiguity under sampling alternatives.	Vessel assay/lineage errors; itinerary-weighted detection and between-vessel variation.	Lineage/assay truth tests, dose/fitness qualification and source identification; then bounded confirmation.

P5 response: report true infection/illness effects together with isolation, staffing, access and other burdens. Verify policy state transitions and event order. A decrease in reported cases alone is not evidence of prevention. Include service/caregiver pathways when they change the intervention's mechanism. Edison: 8171e84c

P6 HVAC: package the existing narrow commissioning evidence, including the corrected explanation of earlier discrepancies. Qualify each additional hull before transporting the result. Solver agreement with intended flows supports an engineering claim; infection-effect claims also need appropriate exposure/dose-response evidence and explicit uncertainty in operations. Do not revive the withdrawn solver-shunt explanation or infer all-hull field validation. [7]

P7 fleet–port inference: test itinerary connectivity, design rank and synthetic recovery before expensive hierarchical inference. Source-market, provisioning, port and calendar effects can be inseparable. Use active relevant observation/source fixtures, not a degenerate example that leaves a channel unused. Report only identified contrasts within connected components if global levels are unsupported. Preserve voyage likelihoods and vessel residuals beneath the fleet model. [38, 39]

P8 variants: qualify lineage inheritance, mixture/contamination and assay error using known synthetic truth, then test pathogen-specific dose/fitness and held-out benchmarks. Detection of a variant and inference of its emergence location are different claims. Existing Paper3 designs do not establish completed valid campaigns. Additional source information is required before geographic attribution. [40]

Publication positioning. P1 is a methods/validation article with a possible software companion; P2/P3/P5 are decision analyses, not clinical trials; P4/P6 may fit environmental/engineering audiences; P7/P8 depend on inferential recovery and identification. Before choosing a journal, assess actual results and a focused novelty review. The present source review is not a comprehensive novelty search across all eight modalities. Distinct estimands and decisions—not output-folder count—justify separate papers.

9 Adaptive campaigns and reproducible claim lineage

Organize work around claims and information gain. First query the existing evidence index and classify each candidate run: **reuse as-is**, **rescore in AWS**, **bridge a changed engine/observation contract**, **new simulation**, or **exclude with reason**. An old SHA is not automatically unusable; a current SHA does not retroactively repair an old archive. Preserve exact engine and analysis versions. Edison: `8171e84c`

9.1 Sequential learning, protected confirmation

Exploration should discriminate competing mechanisms, identify sensitive decisions and expose nonidentifiability. Use paired seed blocks, small factorial contrasts or adaptive enrichment where they are informative; do not repeat a rigid grid merely because the previous paper used it. Existing ordering/stop-rule machinery is a starting point, not automatic statistical validity for every estimand. [1]

Before confirmation freeze estimands, target frame, likelihood/priors, comparator, material-effect threshold, seed policy, independent data split and stopping design. Either use an independent pilot to plan a fixed confirmatory sample, or predeclare a valid sequential inference rule with a compute cap. Repeatedly inspecting ordinary 95% intervals and stopping when convenient is not that rule.

For independent paired blocks $D_b = Y_{A,b} - Y_{B,b}$, report MCSE $= s_D/\sqrt{B}$, where MCSE is Monte Carlo standard error. Choose precision in the outcome's units relative to a scientifically material effect, not a universal seed count. Separate seed error from parameter, observation, structural and target-composition uncertainty. Stop when the declared decision/precision goal is met or the cap is reached; report inconclusive precision honestly. The aims, data-generating mechanisms, estimands, methods and performance measures (ADEMP) framework provides a useful reporting checklist. [41]

Retain extinction and no-detection outcomes, failed cells and negative hypotheses. Diagnose software failures separately from biology. Count retries as attempts linked to the same intended replicate, not new independent evidence. Cluster repeated voyages/shared crews/ports appropriately; individual simulated persons are not independent experimental replicates.

9.2 A small but complete evidence graph

Maintain **run** → **analysis** → **claim** → **figure/table** → **paper**. A run record identifies immutable inputs, effective configuration, engine/container/environment, seeds, completion/failure and raw object/member hashes. An analysis record adds reducer/Stan code, units, filters, weights, denominators, uncertainty and output hashes. A claim adds supported target/domain, calibration/validation sources and approval. A figure/table names its exact query and analysis artifact; the paper freezes its claim set.

For fused archives, distinguish the container hash from each uncompressed member and its run identity. Runtime wiring should extend the proposed provenance envelope rather than duplicate it. These records provide scientific identity and reproducibility; they need not become a tamper-proof attestation project. [8]

A revision or reviewer request creates a parent-linked analysis/attempt and an explicit impact assessment. Reuse unaffected claims; bridge changed mechanisms; supersede only identified claim/figure versions. Never silently replace historical reports or call post hoc work the original confirmation. Edison can assemble manuscripts and figures from qualified compact outputs; the author checks scientific interpretation and signs release.

10 Release package, decisions and limitations

10.1 Make reuse possible without a full AWS campaign

Release a documented, supported core with experimental modules clearly labeled. Existing metadata declares an MIT-licensed Python package; that is not proof of a working clean installation. The publication review found no top-level `CITATION.cff`. Build on that baseline rather than propose software from scratch. [42, 43] Edison: `8171e84c`

The acceptance package should include:

- **Install and tutorial:** wheel/source distribution tested outside the checkout; resource and entry-point checks; supported environment; container digest/build recipe; one laptop-scale synthetic vessel example and an explicitly synthetic fleet/design-set mixture.
- **Scientific documentation:** versioned ODD, TRACE crosswalk, parameter provenance and units, process/event order, observation/response definitions, admissibility cards, defaults and known limitations. [34, 35]
- **Small reference data and notebooks:** license-clean inputs, expected outputs/tolerances, conservation and timing tests, null policy and extinct-run cases, malformed-input checks, analysis/plot recipes and selected telemetry. Synthetic recovery is labeled synthetic, not empirical validation.
- **Reproducible paper capsule:** frozen manifest/environment, run and analysis index, claim-to-figure links, compact tables with schemas, uncertainty calculations, failed attempts, raw retention/access policy and selected independent reproduction.
- **Release metadata:** human-approved authors/contributors, `CITATION.cff`, software and data/dependency licenses, versioned tag, release notes and archived code/permitted data with a version digital object identifier (DOI). No DOI has been minted here. Test reuse from the archived artifact. FAIR principles for research software support explicit identification and reuse. [44]

Do not require proprietary integrations or restricted vessel/clinical data for getting started. Share permitted derived products, synthetic facsimiles and access instructions where original data cannot be released. A software paper is a companion to—not a substitute for—the scientific findings. Journal of Open Source Software guidance requires mature research software, documentation/tests and public development history; assess eligibility at submission rather than promise acceptance. [45]

10.2 Decisions Benjamin and the responsible owners need to make

1. Confirm AWS account, region, buckets, compute queues, existing controller/network topology, image registry, IAM roles and current deployment owner. Supply metadata through approved channels, not keys in chat.
2. Approve the budget/admission policy, reservation and cancellation rules, retention/access policy and the minimal qualification scope. Measure before sizing a bulk campaign.
3. Choose the first P1 narrative and priority secondary manuscript; assign disease, measurement, inference and release owners. Define material effects and genuinely independent validation data.
4. Approve both vessel targets and fleet/design-set reporting frames; **do not choose vessel or fleet as mutually exclusive programs**. Identify any named-fleet traffic data needed for broader claims.
5. Authorize one-time secure integrations and confidentiality conditions. Require the unattended result-consumption acceptance test before promising a fully autonomous loop.

10.3 Limits of this decision report

No deployment, paid AWS job, live workflow dispatch, repository commit or new scientific campaign was performed. Account topology, prices, live archive sizes/completion and external callback availability remain unknown. The 260 GB volume is user supplied. Current source and saved documentary evidence were synthesized; ledger measurements were not independently rerun. Some current summaries retain historical diagnoses and conflicting headers. Claim-specific evidence review remains necessary. The integration and effort plans are proposals, and the eight-paper program does not establish novelty, clinical effectiveness or journal acceptance.

Bottom line: automate small requests and validated compact results around AWS-resident evidence; advance a bounded first methods paper and reusable release while preserving both vessel and fleet inference. Authorize the smallest discriminating next step, not another bulk campaign by default.

Edison Platform Data Sources

The four linked source tasks distinguish code inspection, documented interface feasibility, scientific program design and targeted correction. They are not new campaign executions. The archived evidence preserves original source paths and snapshots; its reconciliation memo controls over stale derivative summaries. Primary repository references below use immutable GitHub URLs, so local workspace paths are not the only evidence route.

Analysis	Type	Trajectory
Source tasks		
Current source audit	Analysis	`5d4be453`
AWS control and data architecture	Analysis	`76fa2216`
Publication and model-reuse program	Analysis	`8171e84c`
Current-source reconciliation	Analysis	`f85969ba`

Supporting deliverables

- Source evidence archive.
- Full cited BibTeX.
- Implementation/publication roadmap JSON.

The evidence archive contains `RECONCILIATION.md`, corrected source reports, retained API/check snapshots, the fixed-SHA source tree subset, official documentation snapshots and an archive-level hash inventory. Earlier derived summaries and line maps are retained as historical source material. The machine roadmap is a proposed plan, not an operational manifest. Full BibTeX is supplied for all 45 references; scholarly DOI identities were checked against Crossref.

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